

AAAD SPACE
START JOB INFORMATION
ENGINEER DATE 15-Jun-13
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES

1 6.513 17.363 2.512; 2 6.374 17.363 2.512; 3 3.869 17.363 2.512;
4 3.778 17.363 2.512; 5 6.374 16.615 2.512; 6 6.513 16.575 2.512;
7 6.752 17.363 1.738; 8 6.374 17.363 1.738; 9 3.869 17.363 1.738;
10 3.778 17.363 1.738; 11 6.374 16.675 1.738; 12 6.752 16.575 1.738;
13 6.374 17.363 2.313; 14 6.374 14.538 2.313; 15 6.374 14.238 2.313;
16 6.374 11.738 2.313; 17 6.374 11.438 2.313; 18 6.374 8.938 2.313;
19 6.374 8.438 2.313; 20 6.374 6.138 2.313; 21 6.374 5.438 2.313;
22 6.374 3.338 2.313; 23 6.374 2.438 2.313; 24 6.374 0.538 2.313;
25 6.374 0 2.313; 26 3.869 17.363 2.313; 27 3.869 14.538 2.313;
28 3.869 14.238 2.313; 29 3.869 11.738 2.313; 30 3.869 11.438 2.313;
31 3.869 8.938 2.313; 32 3.869 8.438 2.313; 33 3.869 6.138 2.313;
34 3.869 5.438 2.313; 35 3.869 3.338 2.313; 36 3.869 2.438 2.313;
37 3.869 0.538 2.313; 38 3.869 0 2.313; 39 6.374 17.363 1.938;
40 6.374 14.538 1.938; 41 6.374 14.238 1.938; 42 6.374 11.738 1.938;
43 6.374 11.438 1.938; 44 6.374 8.938 1.938; 45 6.374 8.438 1.938;
46 6.374 6.138 1.938; 47 6.374 5.438 1.938; 48 6.374 3.338 1.938;
49 6.374 2.438 1.938; 50 6.374 0.538 1.938; 51 6.374 0 1.938;
52 3.869 17.363 1.938; 53 3.869 14.538 1.938; 54 3.869 14.238 1.938;
55 3.869 11.738 1.938; 56 3.869 11.438 1.938; 57 3.869 8.938 1.938;
58 3.869 8.438 1.938; 59 3.869 6.138 1.938; 60 3.869 5.438 1.938;
61 3.869 3.338 1.938; 62 3.869 2.438 1.938; 63 3.869 0.538 1.938;
64 3.869 0 1.938; 71 7.804 14.538 1.938; 72 7.804 11.738 1.938;
73 7.804 8.938 1.938; 74 7.804 6.138 1.938; 75 7.804 3.338 1.938;
76 7.804 0.538 1.938; 77 6.752 17.363 -1.738; 78 6.374 17.363 -1.738;
79 3.869 17.363 -1.738; 80 3.778 17.363 -1.738; 81 6.374 16.675 -1.738;
82 6.752 16.575 -1.738; 83 6.6512 17.363 -2.512; 84 6.374 17.363 -2.512;
85 3.869 17.363 -2.512; 86 3.778 17.363 -2.512; 87 6.374 16.615 -2.512;
88 6.513 16.575 -2.512; 89 6.374 17.363 -1.937; 90 6.374 14.538 -1.937;
91 6.374 14.238 -1.937; 92 6.374 11.738 -1.937; 93 6.374 11.438 -1.937;
94 6.374 8.938 -1.937; 95 6.374 8.438 -1.937; 96 6.374 6.138 -1.937;
97 6.374 5.438 -1.937; 98 6.374 3.338 -1.937; 99 6.374 2.438 -1.937;
100 6.374 0.538 -1.937; 101 6.374 0 -1.937; 102 3.869 17.363 -1.937;
103 3.869 14.538 -1.937; 104 3.869 14.238 -1.937; 105 3.869 11.738 -1.937;
106 3.869 11.438 -1.937; 107 3.869 8.938 -1.937; 108 3.869 8.438 -1.937;
109 3.869 6.138 -1.937; 110 3.869 5.438 -1.937; 111 3.869 3.338 -1.937;
112 3.869 2.438 -1.937; 113 3.869 0.538 -1.937; 114 3.869 0 -1.937;
115 6.374 17.363 -2.312; 116 6.374 14.538 -2.312; 117 6.374 14.238 -2.312;
118 6.374 11.738 -2.312; 119 6.374 11.438 -2.312; 120 6.374 8.938 -2.312;
121 6.374 8.438 -2.312; 122 6.374 6.138 -2.312; 123 6.374 5.438 -2.312;
124 6.374 3.338 -2.312; 125 6.374 2.438 -2.312; 126 6.374 0.538 -2.312;
127 6.374 0 -2.312; 128 3.869 17.363 -2.312; 129 3.869 14.538 -2.312;
130 3.869 14.238 -2.312; 131 3.869 11.738 -2.312; 132 3.869 11.438 -2.312;
133 3.869 8.938 -2.312; 134 3.869 8.438 -2.312; 135 3.869 6.138 -2.312;
136 3.869 5.438 -2.312; 137 3.869 3.338 -2.312; 138 3.869 2.438 -2.312;
139 3.869 0.538 -2.312; 140 3.869 0 -2.312; 141 7.694 14.538 -1.937;
142 7.694 11.738 -1.937; 143 7.694 8.938 -1.937; 144 7.694 6.138 -1.937;
145 7.694 3.338 -1.937; 146 7.694 0.538 -1.937;

MEMBER INCIDENCES

1 1 2; 2 2 3; 3 3 4; 4 4 5; 5 5 6; 6 2 5; 7 7 8; 8 8 9; 9 9 10; 10 10 11;
11 11 12; 12 8 11; 13 8 39; 14 39 13; 15 13 2; 16 9 52; 17 52 26; 18 26 3;
19 13 14; 20 14 15; 21 15 16; 22 16 17; 23 17 18; 24 18 19; 25 19 20; 26 20 21;
27 21 22; 28 22 23; 29 23 24; 30 24 25; 31 39 40; 32 40 41; 33 41 42; 34 42 43;
35 43 44; 36 44 45; 37 45 46; 38 46 47; 39 47 48; 40 48 49; 41 49 50; 42 50 51;
43 43 26; 44 27 28; 45 28 29; 46 29 30; 47 30 31; 48 31 32; 49 32 33; 50 33 34;
51 34 35; 52 35 36; 53 36 37; 54 37 38; 55 52 53; 56 53 54; 57 54 55; 58 55 56;
59 56 57; 60 57 58; 61 58 59; 62 59 60; 63 60 61; 64 61 62; 65 62 63; 66 63 64;
67 40 14; 68 53 27; 69 40 53; 70 14 27; 71 42 16; 72 55 29; 73 42 55; 74 16 29;
75 44 18; 76 57 31; 77 44 57; 78 18 31; 79 46 20; 80 59 33; 81 46 59; 82 20 33;
83 48 22; 84 61 35; 85 48 61; 86 22 35; 87 50 24; 88 63 37; 89 50 63; 90 24 37;
92 40 71; 94 42 72; 96 44 73; 98 46 74; 100 48 75; 102 50 76; 103 77 78;
104 78 79; 105 79 80; 106 80 81; 107 81 82; 108 78 81; 109 83 84; 110 84 85;
111 85 86; 112 86 87; 113 87 88; 114 84 87; 115 84 115; 116 115 89; 117 89 78;
118 85 128; 119 128 102; 120 102 79; 121 89 90; 122 90 91; 123 91 92;
124 92 93; 125 93 94; 126 94 95; 127 95 96; 128 96 97; 129 97 98; 130 98 99;
131 99 100; 132 100 101; 133 115 116; 134 116 117; 135 117 118; 136 118 119;
137 119 120; 138 120 121; 139 121 122; 140 122 123; 141 123 124; 142 124 125;
143 125 126; 144 126 127; 145 102 103; 146 103 104; 147 104 105; 148 105 106;
149 106 107; 150 107 108; 151 108 109; 152 109 110; 153 110 111; 154 111 112;
155 112 113; 156 113 114; 157 128 129; 158 129 130; 159 130 131; 160 131 132;
161 132 133; 162 133 134; 163 134 135; 164 135 136; 165 136 137; 166 137 138;
167 138 139; 168 139 140; 169 116 90; 170 129 103; 171 116 129; 172 90 103;
173 118 92; 174 131 105; 175 118 131; 176 92 105; 177 120 94; 178 133 107;
179 120 133; 180 94 107; 181 122 96; 182 135 109; 183 122 135; 184 96 109;
185 124 98; 186 137 111; 187 124 137; 188 98 111; 189 126 100; 190 139 113;
191 126 139; 192 100 113; 193 90 141; 195 92 142; 197 94 143; 199 96 144;
201 98 145; 203 100 146; 205 90 40; 206 103 53; 207 92 42; 208 105 55;
209 94 44; 210 107 57; 211 96 46; 212 109 59; 213 98 48; 214 111 61;
215 100 50; 216 113 63; 217 90 71; 218 141 40; 219 100 76; 220 146 50;
DEFINE MATERIAL START

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ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
END DEFINE MATERIAL
MEMBER PROPERTY BRITISH
**** BRACKETS
1 TO 3 7 TO 9 TABLE ST UB305X127X42
103 TO 105 109 TO 111 TABLE ST UB305X127X42
*** BRACKET TRUSS
4 TO 6 10 TO 12 106 TO 108 112 TO 114 TABLE ST UC152X152X30
*** MAIN BEAMS
*13 TO 18 115 TO 120 TABLE ST UB356X171X45
13 TO 18 115 TO 120 TABLE ST UB305X127X42
*** cols
19 TO 66 121 TO 168 TABLE ST UC152X152X30
217 TO 220 TABLE ST UA50X50X4
MEMBER PROPERTY JAPANESE
*** H beams
67 TO 90 92 94 96 98 100 102 169 TO 193 195 197 199 201 203 205 TO 215 -
216 TABLE ST H100X50X5
CONSTANTS
MATERIAL STEEL ALL
*MEMBER RELEASE
*205 TO 216 START MY MZ
*205 TO 216 END MY MZ
*67 68 71 72 75 76 79 80 83 84 87 88 START MY MZ
*67 68 71 72 75 76 79 80 83 84 87 88 END MY MZ
*MEMBER COMPRESSION
*92 94 96 98 100 102 193 195 197 199 201 203
MEMBER TENSION
217 TO 220
SUPPORTS
1 6 7 12 77 82 83 88 FIXED
*65 TO 76 141 TO 152 PINNED
71 TO 76 141 TO 146 ENFORCED BUT FY FZ MX MY MZ
LOAD 1 LOADTYPE None TITLE DL
SELFWEIGHT Y -1
LOAD 2 LOADTYPE None TITLE LL
MEMBER LOAD
2 8 73 74 81 82 89 90 171 172 179 180 187 188 CON GY -1.769 2.373
2 8 73 74 81 82 89 90 171 172 179 180 187 188 CON GY -1.769 1.298
2 8 73 74 81 82 89 90 171 172 179 180 187 188 CON GY -1.769 0.133
2 8 73 74 81 82 89 90 171 172 179 180 187 188 CON GY -1.769 1.208
LOAD 3 LOADTYPE None TITLE WLX
MEMBER LOAD
69 TO 90 UNI GX -0.0003
113 TO 134 UNI GX 0.0003
LOAD 4 LOADTYPE None TITLE WLZ
MEMBER LOAD
69 TO 90 UNI GZ 0.0003
113 TO 134 UNI GZ 0.0003
LOAD 5 LOADTYPE None TITLE NLX
SELFWEIGHT X -0.015
LOAD 6 LOADTYPE None TITLE NLZ
SELFWEIGHT Z -0.015
*SLS
LOAD COMB 7 1.0DL + 1.0LL
1 1.0 2 1.0
LOAD COMB 8 1.0DL + 1.0LL+1.0WLX
1 1.0 2 1.0 3 1.0
LOAD COMB 9 1.0DL + 1.0LL + 1.0WLZ
1 1.0 2 1.0 4 1.0
LOAD COMB 10 1.0DL + 1.0LL + 1.0NLX
1 1.0 2 1.0 5 1.0
LOAD COMB 11 1.0DL + 1.0LL + 1.0NLZ
1 1.0 2 1.0 6 1.0
*ULS
LOAD COMB 12 1.4DL + 1.6LL
1 1.4 2 1.6
LOAD COMB 13 1.2DL + 1.2LL + 1.2WLX
1 1.2 2 1.2 3 1.2
LOAD COMB 14 1.2DL + 1.2LL + 1.2WLZ
1 1.2 2 1.2 4 1.2
LOAD COMB 15 1.2DL + 1.2LL + 1.2NLX
1 1.2 2 1.2 5 1.2
LOAD COMB 16 1.2DL + 1.2LL + 1.2NLZ
1 1.2 2 1.2 6 1.2

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PERFORM ANALYSIS
*****
LOAD LIST 7 TO 11
PRINT JOINT DISPLACEMENTS ALL
*****
PRINT SUPPORT REACTION ALL
*****
LOAD LIST 12 TO 16
*****
LOAD LIST ALL
PARAMETER 1
CODE BS5950
LY 3 MEMB 1 TO 3 7 TO 9 13 TO 66 103 TO 105 109 TO 111 115 TO 168
LZ 3 MEMB 1 TO 3 7 TO 9 13 TO 66 103 TO 105 109 TO 111 115 TO 168
BEAM 2 MEMB 1 TO 90 92 94 96 98 100 102 TO 193 195 197 199 201 203 205 TO 216
TRACK 2 MEMB 1 TO 3 7 TO 9 13 TO 66 115 TO 168
TRACK 4 MEMB 1 TO 3 7 TO 9 13 TO 66 115 TO 168
DFF 8 MEMB 1 TO 3 7 TO 9 13 TO 66 115 TO 168
CHECK CODE MEMB 1 TO 90 92 94 96 98 100 102 TO 193 195 197 199 201 203 205 -
206 TO 216
STEEL TAKE OFF LIST 1 TO 90 92 94 96 98 100 102 TO 193 195 197 199 201 203 -
205 TO 216
*****
FINISH
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